

Quantum Mechanics Bootcamp

Ultimately we want a quantum mechanical description of particles & their interactions.

Before introducing quantum fields, it will be useful to review the basics of non-relativistic quantum mechanics.

We'll start with the "shut up & calculate" postulates of quantum mechanics, & eventually get to non-relativistic scattering. From there, we will be prepared to introduce Fermi's Golden rule, particle states, symmetries, and then: QFT.

The postulates of Quantum Mechanics:

Consider a single degree of freedom
(e.g., a particle moving in one dimension)

Postulate 1: The state of the system is described by a vector $|\psi(t)\rangle$ in a Hilbert space
(normed inner product space)

Postulate 2: Observable quantities are represented by Hermitian operators.

E.g., classical dynamical variables x, p are promoted to Hermitian operators X and P which satisfy the canonical commutation relations:

$$[X, P] = i\hbar$$

Postulate 3: If the particle is in a state $|\Psi\rangle$, performing a "measurement" of an observable $\hat{Q}$ will return one of the eigenvalues ω of the operator $\hat{Q}$ with probability:

$$P(\omega) \propto |\langle \omega | \Psi \rangle|^2$$

where $|\omega\rangle$ is the eigenvector: $\hat{Q}|\omega\rangle = \omega|\omega\rangle$

The act of measurement changes the state from $|\Psi\rangle$ to $|\omega\rangle$.

Postulate 4: The time-evolution of the state vector $|\Psi(t)\rangle$ is governed by the Schrödinger equation:

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = H |\Psi(t)\rangle$$

where H is the quantum Hamiltonian operator.

Let's unpack these a bit...

Postulate 1: The state of the system is described by a vector in Hilbert space, $|\Psi(t)\rangle$

A vector space: set of vectors endowed w/
rules for addition ($|\alpha\rangle + |\beta\rangle$) &
multiplication by scalars (complex numbers)
 $a \in \mathbb{C}$, $a|\alpha\rangle + b|\beta\rangle$

Satisfy:

- closure: $|\alpha\rangle + |\beta\rangle$ must be part of the vector space

- scalar multiplication is distributive

$$a(|\alpha\rangle + |\beta\rangle) = a|\alpha\rangle + a|\beta\rangle$$

$$(a+b)|\alpha\rangle = a|\alpha\rangle + b|\alpha\rangle$$

- multiplication is associative:

$$a(b|\alpha\rangle) = ab|\alpha\rangle = b(a|\alpha\rangle)$$

- vector addition is commutative:

$$|\alpha\rangle + |\beta\rangle = |\beta\rangle + |\alpha\rangle$$

- vector addition is associative:

$$|\alpha\rangle + (|\beta\rangle + |\gamma\rangle) = (|\alpha\rangle + |\beta\rangle) + |\gamma\rangle$$

- There exists a null vector $|0\rangle$ such that

$$|\alpha\rangle + |0\rangle = |\alpha\rangle$$

- every vector has an inverse, $|- \alpha\rangle$ s.t.

$$|\alpha\rangle + |- \alpha\rangle = |0\rangle.$$

An inner product is a function that maps two vectors to ~~vectors~~ a scalar, and is denoted:

$$\langle \beta | \alpha \rangle$$

it has the following properties:

$$- \langle \beta | \alpha \rangle = \langle \alpha | \beta \rangle^*$$

$$- \langle \alpha | \alpha \rangle \geq 0 \text{ and } \langle \alpha | \alpha \rangle = 0 \Leftrightarrow |\alpha\rangle = |0\rangle$$

$$- \langle \alpha | (b|\beta\rangle + c|\gamma\rangle) = b\langle \alpha | \beta \rangle + c\langle \alpha | \gamma \rangle$$

A familiar example is given by ordinary vectors in Euclidean space, w/ the dot product as the inner product.

Hilbert spaces: generalization to vector spaces that are complex & even infinite-dimensional.

Aside: "bra" "ket" notation

"ket" vectors denoted $|\alpha\rangle, |\beta\rangle, \dots$

"bra" vectors are the "adjoints", ~~mem~~ elements of the "dual space", $\langle\alpha|, \langle\beta|$

Operationally, adjoint operation applied to a complex vector is given by complex conjugation + transposition (column $\rightarrow$ row)

Inner product of $|\alpha\rangle, |\beta\rangle$ obtained by taking adjoint, and forming a "bra"-ket $\langle\beta|\alpha\rangle$.

It's always possible to find ~~an~~ orthonormal basis of vectors, $\{|e_1\rangle, |e_2\rangle, \dots\}$

which are orthogonal & normalized to unity.

$$\langle e_i | e_j \rangle = \delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

(6)

The basis vectors also satisfy a completeness relation:

$$\sum_i |e_i\rangle\langle e_i| = \mathbb{1} \quad (\text{identity operator})$$

We can expand any vector in terms of an orthonormal basis w/ this:

$$|\alpha\rangle = \mathbb{1}|\alpha\rangle = \sum_i |e_i\rangle \underbrace{\langle e_i|\alpha\rangle}_{\equiv \alpha_i}$$

$$\Rightarrow |\alpha\rangle = \sum_i \alpha_i |e_i\rangle$$

$$= \alpha_1 |e_1\rangle + \alpha_2 |e_2\rangle + \dots$$

The coefficients are $\alpha_i = \langle e_i|\alpha\rangle$.

As an example, return to the case of Euclidean space, unit vectors form a basis: $\hat{i} \hat{j} \hat{k}$

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1,$$

$$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

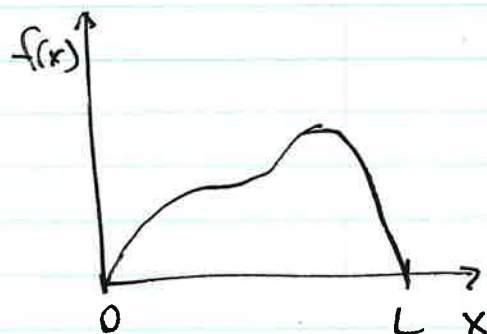
$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Infinite dimensional Hilbert spaces

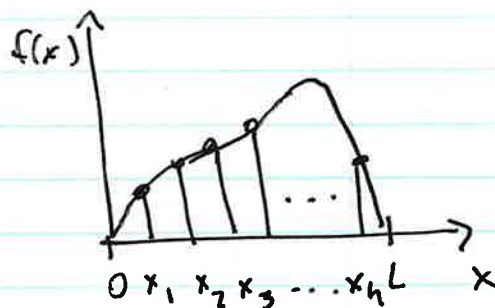
It will be useful to ~~can~~ build up some intuition for infinite-dimensional Hilbert spaces, as we'll be studying them in great detail.

Consider an example of a string with endpoints fixed at $x=0$ and $x=L$.

Suppose at a given time its displacement is given by $f(x)$:



We could approximately describe the displacement by dividing the interval into $n+1$ segments & specifying the displacement at each $x_1, x_2, \dots$



$$\text{by } f_{(n)}(x_1) = f(x_1)$$

$$f_{(n)}(x_2) = f(x_2), \dots, f_{(n)}(x_n) = f(x_n)$$

$\Rightarrow$ interpret these n displacements as ~~as a set~~ components of a ket $|f_{(n)}\rangle$ in an n -dimensional Hilbert space

$$|f_{(n)}\rangle \rightarrow \begin{bmatrix} f_{(n)}(x_1) \\ f_{(n)}(x_2) \\ \vdots \\ f_{(n)}(x_n) \end{bmatrix}$$

We can write the orthonormal basis vectors in this space as $|x_i\rangle$ w/ $i=1, \dots, n$. In components:

$$|x_1\rangle \rightarrow \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, |x_2\rangle \rightarrow \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, |x_n\rangle \rightarrow \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

Clearly, these satisfy: $\langle x_i | x_j \rangle = \delta_{ij}$
(orthonormality)

$$\sum_{i=1}^n |x_i\rangle \langle x_i| = \mathbb{1} \quad (\text{completeness})$$

The ket vector $|f_n\rangle$ is expressed in terms of the basis vectors as:

$$\begin{aligned} |f_n\rangle &= \mathbb{1} |f_n\rangle = \sum_{i=1}^n |x_i\rangle \underbrace{\langle x_i | f_n \rangle}_{f_n(x_i)} \\ &= \sum_{i=1}^n f_n(x_i) |x_i\rangle \end{aligned}$$

We could imagine having different shaped displacements, described by functions $g(x)$, $h(x)$,...

w/ discrete approximations $g_n(x)$, $h_n(x)$,...

a corresponding kets: $|g_n\rangle$, $|h_n\rangle$

The natural inner product is:

$$\begin{aligned}
 \langle f_n | g_n \rangle &= \langle f_n | \mathbb{1} \cdot \mathbb{1} | g_n \rangle \\
 \xrightarrow{\text{(completeness)}} &= \sum_{i=1}^n \sum_{j=1}^n \underbrace{(\langle f_n | x_i \rangle \langle x_i | x_j \rangle \langle x_j | g_n \rangle)}_{\langle x_i | x_j \rangle = \delta_{ij}} \\
 &= \sum_{i=1}^n \sum_{j=1}^n f_n^*(x_i) g_n(x_j) \delta_{ij} \\
 &= \sum_{i=1}^n f_n^*(x_i) g_n(x_i) \quad \swarrow \text{eats up the sum over } j.
 \end{aligned}$$

Now we can take the continuum limit: specify displacement at every point in the interval, represent this as a ket $|f\rangle$
 $\rightarrow$ ordered infinity of values $f(x)$, w/ x a "continuous index".

In going from discrete to continuous, sums are replaced by integrals & the Kronecker delta gets replaced by the Dirac delta function.

The basis vectors are $|x\rangle$ w/ x a continuous index, & these satisfy:

$$\langle x | x' \rangle = \delta(x - x')$$

The Dirac δ -function is zero if $x \neq x'$, and satisfies:

$$\int F(x') \delta(x-x') dx' = F(x)$$

The basis kets also satisfy the completeness relation:

$$\int dx |x\rangle \langle x| = \mathbb{1}$$

w/ the integral performed over the whole space (0 to L)

We can write $|f\rangle$ in terms of the basis vectors:

$$\begin{aligned} |f\rangle &= \mathbb{1} |f\rangle = \int dx' |x'\rangle \langle x'| f\rangle \\ &= \int f(x') |x'\rangle dx' \end{aligned}$$

The function $f(x)$ represents the "component" of the ket $|f\rangle$ along the $|x\rangle$ direction, and is obtained by projecting $|f\rangle$ along $|x\rangle$:

$$\begin{aligned} \langle x|f\rangle &= \int f(x') \langle x|x'\rangle dx' \\ &= \int f(x') \delta(x'-x) dx' = f(x) \end{aligned}$$

The inner product is written as

$$\begin{aligned}
 \langle f | g \rangle &= \langle f | \mathbb{1} \cdot \mathbb{1} | g \rangle \\
 &= \int \langle f | x \rangle \langle x | x' \rangle \langle x' | g \rangle dx dx' \\
 &= \int f^*(x) g(x') \delta(x - x') dx dx' \\
 &= \int f^*(x) g(x) dx
 \end{aligned}$$

Returning to postulate 1: the state ket $|\Psi(t)\rangle$ can be written in the $|x\rangle$ basis as

$$\begin{aligned}
 |\Psi(t)\rangle &= \mathbb{1} |\Psi(t)\rangle = \int |x'\rangle \langle x' | \Psi(t) \rangle dx' \\
 &= \int \Psi(x', t) |x'\rangle dx'
 \end{aligned}$$

Projecting along $|x\rangle$, we have

$$\begin{aligned}
 \langle x | \Psi(t) \rangle &= \int \Psi(x', t) \langle x | x' \rangle dx' \\
 &= \int \Psi(x', t) \delta(x - x') dx' = \Psi(x, t)
 \end{aligned}$$

This is the (spatial) wavefunction of the state $|\Psi\rangle$.

Moving on:

Postulate 2: Observable quantities are represented by Hermitian operators.

E.g., x, p promoted to X, P which satisfy:

$$[X, P] = i\hbar$$

Before discussing observables, let's discuss the "adjoint" operation.

We'll denote the adjoint, or "Hermitian conjugation" w/ a dagger superscript: † .

The adjoint operation on a ket is a bra:

$$(|\alpha\rangle)^\dagger \equiv \langle\alpha|$$

For a complex-valued column-vector, Hermitian conjugation is complex conjugation + transposition.

~~Ann~~

An operator is a mapping Ω of a vector into another vector: $\Omega|\alpha\rangle = |\Omega\alpha\rangle$

$$|\alpha\rangle \rightarrow |\alpha'\rangle = \Omega|\alpha\rangle = |\Omega\alpha\rangle$$

The corresponding mapping of the bra vector is found by taking the adjoint:

$$(|\alpha\rangle)^{\dagger} \rightarrow (|\alpha'\rangle)^{\dagger} = (\Omega|\alpha\rangle)^{\dagger} = (|\Omega\alpha\rangle)^{\dagger}$$

$$\Rightarrow \langle\alpha| \rightarrow \langle\alpha'| = \langle\alpha|\Omega^{\dagger} = \langle\Omega\alpha|$$

i.e., if Ω maps $|\alpha\rangle$ to $|\alpha'\rangle$, $\Omega^{\dagger}$ maps $\langle\alpha|$ to $\langle\alpha'|$.

If we represent the operator in terms of its components in a particular basis, we can group these into a (perhaps infinite dimensional) matrix.

Then, the adjoint is just the conjugate-transpose.

Now postulate 2 distinguished observables as Hermitian operators, i.e., those that satisfy:

$$\Omega^{\dagger} = \Omega \quad (\text{Hermitian})$$

This is a "reality" condition on operators. An important property of Hermitian operators is that their eigenvalues are real.

Before showing this, consider the general eigenvalue problem. Consider $\mathcal{R}$ (not necc. Hermitian) acting on a ket $\mathcal{R}|\alpha\rangle = |\alpha'\rangle$.

An eigenvector (eigenket) $|\omega\rangle$ of $\mathcal{R}$ is one which satisfies

$$\mathcal{R}|\omega\rangle = \omega|\omega\rangle$$

$\mathcal{R}$ same state, rescaled by ω number.

The value ω is the "eigenvalue" associated w/ eigenket $|\omega\rangle$.

To solve for the eigenvalues/eigenstates, we write this as

$$(\mathcal{R} - \omega \mathbb{1})|\omega\rangle = 0$$

$\Rightarrow$ solutions are where

$$\det(\mathcal{R} - \omega \mathbb{1}) \quad \text{"characteristic equation"}$$

$\mathcal{P}$

generalization of matrix determinant to linear operators.

For a Hermitian operator, $\Omega^\dagger = \Omega$,

$$\Omega|\psi\rangle = \omega|\psi\rangle \Rightarrow \langle\psi|\Omega|\psi\rangle = \omega\langle\psi|\psi\rangle$$

taking the adjoint:

$$\langle\psi|\Omega^\dagger|\psi\rangle = \omega^*\langle\psi|\psi\rangle$$

$$\langle\psi|\Omega|\psi\rangle \quad \text{(Hermiticity)}$$

$$\Rightarrow 0 = (\omega - \omega^*)\langle\psi|\psi\rangle$$

$\uparrow$
non-zero, so $\omega = \omega^*$!

In other words, ω is real if Ω is Hermitian
(a good property to have for "observables")

The orthonormal eigenvectors of a Hermitian operator
furnish a basis for the Hilbert space.

The operator is diagonal in this basis, w/ diagonal
elements given by the eigenvalues.

Let's return to P & X . As we stated, these
satisfy:

$$[X, P] = i\hbar \quad (*)$$

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We can use this to establish a representation of P in the X -eigenbasis, labelled by $|x\rangle$.

From (*), we get:

$$[X, P] |\Psi\rangle = i\hbar |\Psi\rangle$$

$$= (XP - PX) |\Psi\rangle = i\hbar |\Psi\rangle.$$

$$\Rightarrow \langle x | (XP - PX) \int |x'\rangle \langle x' | \Psi \rangle dx' = i\hbar |\Psi\rangle$$

$\uparrow$
 $= \mathbb{I}$

now, ~~$\langle x | x \rangle = \delta(x)$~~ and ~~$\langle x | x \rangle = \delta(x)$~~

$$X|x'\rangle = x'|x'\rangle \quad \& \quad \langle x | X = \langle x | x$$

since these are eigenstates, so:

$$\int dx' \left[x \langle x | P | x' \rangle - \langle x | P | x' \rangle x' \right] \langle x' | \Psi \rangle$$

$$= i\hbar \Psi(x) \quad \equiv \text{ ~~$\langle x | P | x \rangle$~~ }$$

$$\text{Or, } \int \text{ ~~$\langle x | P | x' \rangle$~~ } \Psi(x') dx' = i\hbar \Psi(x)$$

$$\langle x | (XP - PX) | x' \rangle$$

For this to hold for arbitrary $|\Psi\rangle$, the matrix element must be proportional to the Dirac δ -function:

$$\langle x|P|x'\rangle = \delta(x-x') P(x')$$

$\uparrow$
operator in position space
to be determined!

Then,

$$\int dx' \left(x \delta(x-x') P(x') \Psi(x') - P(x') x' \Psi(x') \delta(x-x') \right) = i\hbar \Psi(x)$$

or,

$$x P(x) \Psi(x) - P(x) x \Psi(x) = i\hbar \Psi(x)$$

The solution of this is $\boxed{P(x) = -i\hbar \frac{d}{dx}}$

$$-i\hbar x \frac{d\Psi}{dx} + i\hbar \frac{d}{dx}(x\Psi(x)) = i\hbar \Psi(x)$$

So in the X -basis, we have the representation:

$$|\Psi\rangle \rightarrow \Psi(x)$$

$$X \rightarrow x$$

$$P \rightarrow -i\hbar \frac{d}{dx}$$

Postulate 3: If the particle is in a state $|\Phi\rangle$, a measurement of the observable Ω will return one of the eigenvalues ω_i w/ probability

$$P(\omega_i) \propto |\langle \omega_i | \Phi \rangle|^2$$

(where $\Omega |\omega_i\rangle = \omega_i |\omega_i\rangle$)

The act of measurement changes the state to $|\omega_i\rangle$.

This allows us to make statistical predictions.

Recall that the orthonormal eigenvectors of Ω constitute a complete basis for the Hilbert space. Denote these eigenvectors

$$\{|\omega_1\rangle, |\omega_2\rangle, \dots\}$$

w/ corresponding eigenvalues $\{\omega_1, \omega_2, \dots\}$

We've already discussed how to find these eigenvalues / eigenvectors.

Now, we expand $|\Phi\rangle$ in the Ω basis:

$$|\Phi\rangle = \mathbb{I} |\Phi\rangle$$

$$= \sum_j |\omega_j\rangle \langle \omega_j | \Phi \rangle \equiv \sum_j |\omega_j\rangle \varphi_j$$

$$= \varphi_1 |\omega_1\rangle + \varphi_2 |\omega_2\rangle + \dots$$

We see that $|\Psi\rangle$ is in a superposition of Ω eigenstates.

The third postulate states that upon measuring Ω , the result will be one of these eigenvalues, and the probability of each measurement is

$$P(\omega_i) \propto |\langle \omega_i | \Psi \rangle|^2 \equiv |\varphi_i|^2$$

$\uparrow$ probability amplitude

$\rightarrow$ projection of $|\Psi\rangle$ along $|\omega_i\rangle$.

To find the absolute probability, we divide by the sum of all relative probabilities:

$$\begin{aligned} P(\omega_i) &= \frac{|\langle \omega_i | \Psi \rangle|^2}{\sum_j |\langle \omega_j | \Psi \rangle|^2} = \frac{|\langle \omega_i | \Psi \rangle|^2}{\sum_j \langle \Psi | \omega_j \rangle \langle \omega_j | \Psi \rangle} \\ &= \frac{|\langle \omega_i | \Psi \rangle|^2}{\langle \Psi | \mathbb{I} | \Psi \rangle} = \frac{|\langle \omega_i | \Psi \rangle|^2}{\langle \Psi | \Psi \rangle} \end{aligned}$$

$\Rightarrow$ if we had instead used normalized states,

$$|\Psi'\rangle = \frac{|\Psi\rangle}{\sqrt{\langle \Psi | \Psi \rangle}}$$

The probability would be simply

$$P(\omega_i) = |\langle \omega_i | \Psi \rangle|^2$$

How does this work in the infinite-dimensional case?

Consider again a particle moving in 1d described by a ket $|\Psi\rangle$. Suppose $|\Psi\rangle$ is normalized, i.e.,

$$\langle \Psi | \Psi \rangle = 1.$$

We can expand $|\Psi\rangle$ in the x -basis as:

$$\begin{aligned} |\Psi\rangle &= \mathbb{I} |\Psi\rangle = \int |x\rangle \langle x | \Psi \rangle dx \\ &= \int |x\rangle \Psi(x) dx \end{aligned}$$

As in the finite dimensional case, we expect

$$\langle x | \Psi \rangle = \Psi(x) \text{ (the wavefunction) to tell}$$

us something about a position measurement.

To get the correct interpretation, again use the complete set of states:

$$\begin{aligned} 1 &= \langle \Psi | \Psi \rangle = \langle \Psi | \mathbb{I} | \Psi \rangle = \int \langle \Psi | x \rangle \langle x | \Psi \rangle dx \\ &= \int \Psi^*(x) \Psi(x) dx = \int |\Psi(x)|^2 dx \end{aligned}$$

This is the familiar idea that $|\Psi(x)|^2 = |\langle x | \Psi \rangle|^2$ is the probability density, i.e., the probability of measuring the particle between x_1 and x_2 is:

$$P(x_1 \leq x \leq x_2) = \int_{x_1}^{x_2} |\Psi(x)|^2 dx$$

Expectation values and uncertainty

Suppose we have a large ensemble of particles in a state $|\Psi\rangle$. We can perform a large number of measurements of an observable, Ω and then characterize the outcome in terms of statistical measures, such as the mean and variance.

The expectation value $\langle \Omega \rangle$ is just the mean.

In the finite dimensional case, if w_i, w_j are normalized eigenvectors of Ω ($\Omega |w_i\rangle = w_i |w_i\rangle$)

Then the expectation value is just

$$\begin{aligned} \langle \Omega \rangle &= \sum_i P(w_i) w_i = \sum_i |\langle w_i | \Psi \rangle|^2 w_i \\ &= \sum_i \langle \Psi | w_i \rangle \langle w_i | \Psi \rangle w_i \end{aligned}$$

$$\begin{aligned}
 &= \sum_i \langle \Psi | \psi_i | \psi_i \rangle \langle \psi_i | \Psi \rangle \\
 &= \sum_i \langle \Psi | \Omega | \psi_i \rangle \langle \psi_i | \Psi \rangle \\
 &\stackrel{\text{completeness relation}}{=} \langle \Psi | \Omega | \Psi \rangle \quad \swarrow \text{since } \psi_i | \psi_i \rangle = \Omega | \psi_i \rangle
 \end{aligned}$$

$\Rightarrow$ last relation holds in any basis
(if we know representation of Ω)

Similarly, we can compute the uncertainty which is just the standard deviation (measures the size of average fluctuations about the mean)

$$\Delta \Omega = \left\langle (\Omega - \langle \Omega \rangle)^2 \right\rangle^{1/2}$$

A similar analysis as above yields

$$\Delta \Omega = \left[\langle \Psi | (\Omega - \langle \Omega \rangle)^2 | \Psi \rangle \right]^{1/2}$$

For two operators respecting canonical commutation relations $[\Omega, \Lambda] = i\hbar$, such as X, P , the products of their uncertainties satisfy

$$\Delta \Omega \Delta \Lambda \geq \frac{\hbar}{2} \quad \text{— Heisenberg uncertainty relation.}$$

Postulate 4: the time evolution of the state $|\Psi(t)\rangle$ is governed by the Hamiltonian operator via the Schrödinger equation:

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

For a particle moving in 1d, we can obtain the Hamiltonian H from "quantizing" the classical Hamiltonian; i.e., by substituting

$$x \rightarrow \hat{X}, \quad p \rightarrow \hat{P} \quad \left(\hat{} \text{ indicates operators for emphasis} \right)$$

Ex. 1-dim quantum harmonic oscillator has classical potential $V(x) = \frac{1}{2} m \omega^2 x^2$

$$\Rightarrow \hat{H} = \frac{1}{2m} \hat{P}^2 + \frac{1}{2} m \omega^2 \hat{X}^2$$

In general there may be additional terms that don't depend on X, P (e.g., spin)

Consider the situation where H has no explicit time-dependence. Given H , we can find the energy eigenvalues E_i , & eigenstates $|E_i\rangle$ satisfying:

$$H|E_i\rangle = E_i|E_i\rangle$$

→ "time-independent Schrödinger equation"

The $|E_i\rangle$ form a complete basis for the Hilbert space, so we can expand $|\Psi(t)\rangle$ as:

$$\begin{aligned} |\Psi(t)\rangle &= \sum_i |E_i\rangle \langle E_i | \Psi(t) \rangle \\ &= \sum_i c_E^i(t) |E_i\rangle \end{aligned}$$

Now plug this into the time-dependent Schrödinger equation:

$$\begin{aligned} i\hbar \frac{d}{dt} \left[\sum_i c_E^i(t) |E_i\rangle \right] &= \hat{H} \left[\sum_i c_E^i(t) |E_i\rangle \right] \\ \Downarrow &\quad \quad \quad \Downarrow \\ = \sum_i \left[i\hbar \frac{dc_E^i(t)}{dt} |E_i\rangle \right] &= \sum_i c_E^i(t) E_i |E_i\rangle \end{aligned}$$

$$\Rightarrow \sum_i \left[i\hbar \frac{d\varphi_E^i(t)}{dt} - E_i \varphi_E^i(t) \right] |E_i\rangle = 0$$

or just:
$$i\hbar \frac{d\varphi_E^i(t)}{dt} - E_i \varphi_E^i(t) = 0$$

The solution to this is

$$\varphi_E^i(t) = \varphi_E^i(0) e^{-iE_i t/\hbar}$$

or equivalently,

$$\langle E_i | \varphi(t) \rangle = \langle E_i | \varphi(0) \rangle e^{-iE_i t/\hbar}$$

If we plug this back into our expansion of $|\varphi(t)\rangle$ in terms of energy eigenstates:

$$|\varphi(t)\rangle = \sum_i |E_i\rangle \langle E_i | \varphi(0) \rangle e^{-iE_i t/\hbar}$$

$$= \left[\sum_i |E_i\rangle \langle E_i | e^{-iE_i t/\hbar} \right] |\varphi(0)\rangle$$

this is an operator (expanded in the E basis)

→ operator acting on initial state, known as the time evolution operator, or propagator $U(t)$

where

$$U(t) = \sum_i |E_i\rangle \langle E_i| e^{-iE_i t/\hbar}$$

Note that $U(0) = \sum_i |E_i\rangle \langle E_i| = \mathbb{I}$.

Thus, another way of writing the solution to the Schrödinger equation is

$$|\Psi(t)\rangle = U(t) |\Psi(0)\rangle$$

where we can express $U(t)$ directly in terms of H :

$$U(t) = \exp(-i\hat{H}t/\hbar)$$

so that

$$|\Psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\Psi(0)\rangle$$

This suggests we view the time-evolution of the state $|\Psi(t)\rangle$ as a "rotation" in Hilbert space.

Non-relativistic Perturbation Theory

Often the actual Hamiltonian of the system is too complicated to solve explicitly (this will certainly be true in interesting QFTs we study shortly) — however, if the "complicated part" of the Hamiltonian is "small", we can treat this as a perturbation and solve the system order by order in some small parameter. This strategy is known as perturbation theory.

Perturbation theory will be our primary tool for studying QFT & the standard model.

It's worthwhile to orient ourselves in non-relativistic QM systems first; the concepts will generalize easily later.

Consider a Hamiltonian H which can be factorized as

$$H = H_0 + V(\vec{x}, t)$$

"free"
(solvable)
Hamiltonian

"small"
perturbation

Now the $|u_n\rangle$ form a complete basis of the Hilbert space — even in the presence of V ! — so we can expand a solution of the full Hamiltonian:

$$|\Psi(t)\rangle = \sum_n a_n(t) e^{-iE_n t/\hbar} |u_n\rangle$$

Now the Schrödinger equation tells us that

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = H |\Psi(t)\rangle$$

$$\begin{aligned} \Rightarrow \sum_n \left(i\hbar \frac{da_n}{dt} + \cancel{a_n E_n} \right) e^{-iE_n t/\hbar} |u_n\rangle \\ = \cancel{(H_0 + V)} \sum_n a_n(t) e^{-iE_n t/\hbar} |u_n\rangle \end{aligned}$$

But $H_0 |u_n\rangle = E_n |u_n\rangle$ — the $a_n E_n$ term ~~is~~ cancels, and we find:

$$\sum_n \left(i\hbar \frac{da_n}{dt} \right) e^{-iE_n t} |u_n\rangle = \sum_n a_n(t) V e^{-iE_n t} |u_n\rangle$$

Now we contract this w/ a "final state" $\langle u_f |$:

$$\begin{aligned} \langle u_f | \sum_n \left(i\hbar \frac{da_n}{dt} \right) e^{-iE_n t} |u_n\rangle \\ = \sum_n a_n(t) \langle u_f | V e^{-iE_n t} |u_n\rangle \end{aligned}$$

Now, the $|u_i\rangle$ are orthogonal so the LHS becomes:

$$\begin{aligned} \langle u_f | \sum_n \left(i\hbar \frac{da_n}{dt} \right) e^{-iE_n t} | u_n \rangle \\ = \sum_n \left(i\hbar \frac{da_n}{dt} \right) e^{-iE_n t} \underbrace{\langle u_f | u_n \rangle}_{=\delta_{fn}} \\ = i\hbar \frac{da_f}{dt} e^{-iE_f t} \end{aligned}$$

While the right-hand side gives:

$$\begin{aligned} \sum_n a_n(t) e^{-iE_n t / \hbar} \langle u_f | V | u_n \rangle \\ = V_{fn} \quad \text{— a matrix-element} \end{aligned}$$

We find:

$$i\hbar \frac{da_f}{dt} e^{-iE_f t} = \sum_n a_n(t) e^{-iE_n t / \hbar} V_{fn}(t)$$

We contracted w/ $\langle u_f |$ — tells us how "likely" the particle that started in state $|u_n\rangle$ ends in state $\langle u_f |$

This is a complicated, interconnected set of equations, but we can try to solve it order-by-order.

If we assume V_{fi} is small, so that there is only one "transition" between the initial & final states, then $a_n(t)$ on the RHS is ~ 1

$$\rightarrow i\hbar \frac{da_f}{dt} e^{-iE_f t/\hbar} \approx \sum_n 1 \cdot e^{-iE_i t/\hbar} V_{fi} + \mathcal{O}(\alpha)$$

This can easily be solved:

$$a_f(t) \approx -i \int_0^t dt' V_{fi}(t') e^{i(E_f - E_i)t'/\hbar} + \mathcal{O}(\alpha)$$

We can in principle do this to arbitrary order. This first order approximation is known as the Born approximation.

We can make a little more progress by assuming that V is independent of time.

Then,

$$T_{fi} \equiv -i \int_{-\infty}^{\infty} dt' e^{i(E_f - E_i)t'/\hbar} \langle u_f | V | u_i \rangle$$

But if V is t -indep, this comes out of the integral, and

$$T_{fi} = -i V_{fi} \underbrace{\int_{-\infty}^{\infty} dt' e^{+i(E_f - E_i)t'/\hbar}}$$

this is an integral representation of the Dirac δ -function'.

In case this isn't familiar to you, recall that the δ -function is defined so that

$$\int_{-\infty}^{\infty} dx f(x) \delta(x-y) = f(y)$$

~~So, we can~~ But we know from complex analysis that we can rewrite any function as a Fourier transform'.

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk e^{ikx} \tilde{f}(k)$$

where $\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx f(x) e^{-ikx}$

But then:

$$f(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk e^{iky} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx f(x) e^{-ikx} \right)$$

or,

$$f(y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx f(x) \underbrace{\int_{-\infty}^{\infty} dk e^{ik(y-x)}}_{\equiv (2\pi) \delta(x-y)}$$

(We'll use this extensively in field theory later)

Returning to the transition probability:

$$T_{fi} = -i V_{fi} \int_{-\infty}^{\infty} dt' e^{+i(E_f - E_i)t'/\hbar}$$

$$= -i 2\pi \hbar V_{fi} \delta(E_f - E_i)$$

$$= -i 2\pi \hbar V_{fi} \delta(E_f - E_i)$$

→ note that this vanishes when $E_f \neq E_i$!
(Energy conservation)

We've been cavalier about regulating the infinite time limits, but a well-defined quantity is the
"transition probability per unit time"

$$\begin{aligned}
 W_{fi} &\equiv \lim_{T \rightarrow \infty} \frac{|T_{fi}|^2}{T} \\
 &= \lim_{T \rightarrow \infty} \frac{|-i 2\alpha \hbar V_{fi} \delta(E_f - E_i)|^2}{T} \\
 &= \lim_{T \rightarrow \infty} \frac{1}{T} |V_{fi}|^2 (2\alpha)^2 \delta(E_f - E_i) \int_{-T/2}^{T/2} \frac{dt}{2\alpha} e^{+i(E_f - E_i)t/\hbar} \\
 &= \lim_{T \rightarrow \infty} \frac{1}{T} |V_{fi}|^2 2\alpha \delta(E_f - E_i) \underbrace{\int_{-T/2}^{T/2} dt}_{\rightarrow T}
 \end{aligned}$$

$$\Rightarrow W_{fi} = (2\alpha) |V_{fi}|^2 \delta(E_f - E_i)$$

If we assume there is a continuum of ~~states~~ energies E , w/ multiple states sharing the same energy, we can write

$$\rho(E_f) dE_f \equiv \# \text{ of final states w/ energy in range } [E_f, E_f + dE_f]$$

$\Rightarrow$

$$\begin{aligned}
 W_{fi} &= (2\alpha) \int dE_f |V_{fi}|^2 \delta(E_f - E_i) \rho(E_f) \\
 &= (2\alpha) |V_{fi}|^2 \rho(E_f) \quad \left| \text{Fermi's Golden Rule} \right|
 \end{aligned}$$